Where Does the Hedging Error Go? A Greek-by-Greek Decomposition for 0DTE Options under Heston

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Abstract

Zero Days to Expiration Options (0DTE) have increased in popularity since CBOE released daily SPX expirations in 2022. However, they are difficult to hedge, and the sources of hedging error are not well understood. This paper takes the Heston model as a starting point and uses an industry-standard Greek-by-Greek decomposition to analyze the sources of hedging error for a delta hedge rebalanced at five-minute intervals. In this paper, the decomposition attributes the lion's share of error to Gamma, which absorbs both diffusion convexity and any jump risk the model cannot identify. In higher volatility regimes, the attribution share carried by Vega increases from 1% to 11%, while Gamma's falls from 94% to 82%.

1 Introduction

This paper investigates the Greeks as sources of discrete hedging error for 0DTE options under Heston. It takes the standard compensated-Greek decomposition of a delta-hedged option's profit and loss and applies it to at-the-money 0DTE calls, hedged at five-minute frequency on real SPX data. The contribution is a term-by-term attribution of where the hedging error actually goes, and how that attribution shifts with the volatility regime. The headline is that Gamma carries the error in every regime, but the balance between Gamma and Vega swings from roughly 94/1 on the calmest days to 82/11 on the most turbulent.

1.1 Black–Scholes Model and Greeks

The Black–Scholes model, introduced by Fischer Black and Myron Scholes (1973) [1] and Robert Merton (1973) [2], provides a framework for pricing European options in a closed form that Heston expands on. The issue with Black–Scholes is that it assumes constant volatility, which is not the case in real markets, especially for 0DTE options. The Heston model, explained below, allows for stochastic volatility. Black–Scholes also introduced the concept of Greeks, which are sensitivities of the option price to various parameters. They are listed below:

- Delta (Δ): Sensitivity of the option price to changes in the underlying asset price.
- Gamma (Γ): Sensitivity of Delta to changes in the underlying asset price.
- Vega (ν): Sensitivity of the option price to changes in volatility.
- Theta (Θ): Sensitivity of the option price to the passage of time.
- Charm ($\frac{\partial \Delta}{\partial t}$): Sensitivity of Delta to the passage of time.
- Vanna ($\frac{\partial \Delta}{\partial \sigma}$): Sensitivity of Delta to changes in volatility.
- Volga, also called Vomma ($\frac{\partial \nu}{\partial \sigma}$): Sensitivity of Vega to changes in volatility.
- Veta ($\frac{\partial \nu}{\partial t}$): Sensitivity of Vega to the passage of time.
- Time convexity ($\frac{\partial^2 V}{\partial t^2}$): Second-order sensitivity of the option price to the passage of time.

2 Background

2.1 Related Literature

The decomposition in this paper relies heavily on various prior works. Firstly, the Heston model [3] is used as the underlying model for the option pricing and hedging found later in the paper. Along with that, Euler–Maruyama [13] is used to simulate the stochastic differential equations used in the implementation of the paper. The Greek decomposition is based on Bergomi's work [9], which is used to decompose the hedging error into its various components. Specifically, his "break-even" logic is used to attribute the hedging error to each Greek. Bergomi's logic is the implied vs. realized volatility, where dominating volatility leads to price dominating the true price, and the Gamma variance is the engine that drives the hedging error. This implied-versus-realized logic was formalized by El Karoui, Jeanblanc-Picqué, and Shreve (1998) [5], who show that the P&L of a delta hedge run at the wrong volatility is the dollar-Gamma-weighted spread between the hedging and realized variances.

Kamal [6] is the predecessor to this project. His work on Black–Scholes mirrors this paper because it is a practitioner's guide to hedging errors when one cannot hedge continuously. His major discovery is that hedging error is proportional to $\frac{1}{\sqrt{N}}$, where N is the number of times an option has been hedged. Bertsimas–Kogan–Lo [7] is the successor to that paper, and discovers that discrete tracking error grows with the granularity constant, g, which is built from accumulated dollar gamma. The formal contribution of Bertsimas et al. is the expression $\frac{g}{\sqrt{N}}$; they also show that the kink in the call payoff does not break this convergence rate, but rather inflates g. Gobet and Temam (2001) [8] show that the convergence rate depends on the regularity of the payoff, but do not provide any specific constants or claim that the kink degrades the rate for calls. Dim, Eraker, and Vilkov (2023) [10] study how 0DTE order imbalances and the resulting dealer gamma exposure propagate into market volatility—a microstructure channel, distinct from the contract-level, term-by-term attribution pursued here. This paper fills a gap in the literature by providing a term-by-term hedging-error attribution for 0DTE options under Heston.

2.2 Choices and Rationale

The Heston model was chosen for its stochastic volatility and easily implementable characteristic function. Also, the Heston parameters are round, arbitrarily chosen values rather than calibrated estimates, except for η (see the Methodology section for further information). Their specific combination violates the Feller condition, so the variance process can reach zero; the simulation therefore uses a full-truncation Euler–Maruyama scheme, which floors the variance at zero at each step.

2.3 Heston Model Derivation

The derivation here follows both Gatheral (2006) [4] and Heston (1993) [3]. The derivation starts with the Heston model's stochastic differential equations for the underlying asset price S and its variance v:

$$dS = \mu S\,dt + \sqrt{v}\,S\,dX, \qquad dv = \kappa(\eta - v)\,dt + \epsilon\sqrt{v}\,dW, \qquad dX\,dW = \rho\,dt,$$

There are two sources of randomness here, the price shock dX and the variance shock dW, so the hedge takes two instruments: the stock, and a second option V_1 with its own exposure to the variance. Take the portfolio, Π, with the option V by shorting ΔS, which is the dollar value of the stock shorted. Short Δ_1 shares of V_1 as well:

$$\Pi = V - \Delta S - \Delta_1 V_1.$$

Applying Itô's lemma to V gives its change over an instant:

$$dV = \left(\frac{\partial V}{\partial t} + \frac{1}{2}vS^2\frac{\partial^2 V}{\partial S^2} + \rho\epsilon vS\frac{\partial^2 V}{\partial S\,\partial v} + \frac{1}{2}\epsilon^2 v\frac{\partial^2 V}{\partial v^2}\right)dt + \frac{\partial V}{\partial S}\,dS + \frac{\partial V}{\partial v}\,dv,$$

Apply the same expansion to V_1, and substitute both into $d\Pi$ to obtain the following equation:

$$\begin{aligned} d\Pi = {} & \left(\frac{\partial V}{\partial t} + \frac{1}{2}vS^2\frac{\partial^2 V}{\partial S^2} + \rho\epsilon vS\frac{\partial^2 V}{\partial S\,\partial v} + \frac{1}{2}\epsilon^2 v\frac{\partial^2 V}{\partial v^2}\right)dt \\ & - \Delta_1\left(\frac{\partial V_1}{\partial t} + \frac{1}{2}vS^2\frac{\partial^2 V_1}{\partial S^2} + \rho\epsilon vS\frac{\partial^2 V_1}{\partial S\,\partial v} + \frac{1}{2}\epsilon^2 v\frac{\partial^2 V_1}{\partial v^2}\right)dt \\ & + \left(\frac{\partial V}{\partial S} - \Delta_1\frac{\partial V_1}{\partial S} - \Delta\right)dS + \left(\frac{\partial V}{\partial v} - \Delta_1\frac{\partial V_1}{\partial v}\right)dv. \end{aligned}$$

The last line shows how to remove the risk: set Δ_1 to cancel the dv exposure, and Δ to cancel the dS exposure,

$$\Delta_1 = \frac{\partial V/\partial v}{\partial V_1/\partial v}, \qquad \Delta = \frac{\partial V}{\partial S} - \Delta_1\frac{\partial V_1}{\partial S},$$

leaving a riskless portfolio. No-arbitrage then forces Π to earn the risk-free rate, $d\Pi = r\Pi\,dt$. Because the same argument holds for any option, the terms involving V and the terms involving V_1 must each equal a common function of (S, v, t); following Heston, that function is written as $-\nu_{\mathrm{Q}}$:

$$\frac{1}{\partial V/\partial v}\left(\frac{\partial V}{\partial t} + \frac{1}{2}vS^2\frac{\partial^2 V}{\partial S^2} + \rho\epsilon vS\frac{\partial^2 V}{\partial S\,\partial v}\right.$$

$$+\frac{1}{2}\epsilon^2 v\frac{\partial^2 V}{\partial v^2}+rS\frac{\partial V}{\partial S}-rV\Big)=-\nu_{\mathrm{Q}},$$

where

$$\nu_{\mathrm{Q}}\equiv\kappa(\eta-v)-\lambda v$$

is the risk-neutral drift of the variance, with λ the market price of volatility risk. Rearranging gives the Heston pricing PDE:

$$\frac{\partial V}{\partial t}+\frac{1}{2}vS^2\frac{\partial^2 V}{\partial S^2}+\rho\epsilon vS\frac{\partial^2 V}{\partial S\,\partial v}+\frac{1}{2}\epsilon^2 v\frac{\partial^2 V}{\partial v^2}$$

$$+rS\frac{\partial V}{\partial S}+[\kappa(\eta-v)-\lambda v]\frac{\partial V}{\partial v}-rV=0.$$

3 The Error Model

3.1 Derivation

The Error model is derived from the Heston model; the two diverge at the continuous limit. The Heston model assumes continuous hedging; therefore, $\Delta t \to 0$. However, in practice, hedging is done at discrete intervals, which introduces hedging error. To decompose said error, the following model is proposed based on Bergomi's compensated-Greek decomposition, adding the discrete-time terms—charm, veta, and time convexity—that arise from expanding the profit and loss to second order in Δt [9]:

$$\Delta\Pi = \Delta V - \frac{\partial V}{\partial S}\,\Delta S - r\left(V - \frac{\partial V}{\partial S}\,S\right)\Delta t.$$

$$\begin{aligned}\Delta\Pi = {} & \frac{\partial V}{\partial t}\,\Delta t + \frac{\partial V}{\partial S}\,\Delta S + \frac{\partial V}{\partial v}\,\Delta v \\ & + \frac{1}{2}\frac{\partial^2 V}{\partial S^2}\,(\Delta S)^2 + \frac{1}{2}\frac{\partial^2 V}{\partial v^2}\,(\Delta v)^2 + \frac{\partial^2 V}{\partial S\,\partial v}\,\Delta S\,\Delta v \\ & + \frac{\partial^2 V}{\partial S\,\partial t}\,\Delta S\,\Delta t + \frac{\partial^2 V}{\partial v\,\partial t}\,\Delta v\,\Delta t + \frac{1}{2}\frac{\partial^2 V}{\partial t^2}\,(\Delta t)^2 \\ & - \frac{\partial V}{\partial S}\,\Delta S - r\left(V - \frac{\partial V}{\partial S}\,S\right)\Delta t.\end{aligned}$$

$$\frac{\partial V}{\partial t} = rV - rS\,\frac{\partial V}{\partial S} - \frac{1}{2}vS^2\,\frac{\partial^2 V}{\partial S^2} - \rho\epsilon vS\,\frac{\partial^2 V}{\partial S\,\partial v} - \frac{1}{2}\epsilon^2 v\,\frac{\partial^2 V}{\partial v^2} - \nu_{\mathrm{Q}}\,\frac{\partial V}{\partial v},$$

$$\begin{aligned}\Delta\Pi = {} & \frac{1}{2}\frac{\partial^2 V}{\partial S^2}[(\Delta S)^2 - vS^2\,\Delta t] + \frac{1}{2}\frac{\partial^2 V}{\partial v^2}[(\Delta v)^2 - \epsilon^2 v\,\Delta t] \\ & + \frac{\partial^2 V}{\partial S\,\partial v}[\Delta S\,\Delta v - \rho\epsilon vS\,\Delta t] + \frac{\partial V}{\partial v}[\Delta v - \nu_{\mathrm{Q}}\,\Delta t] \\ & + \frac{\partial^2 V}{\partial S\,\partial t}\,\Delta S\,\Delta t + \frac{\partial^2 V}{\partial v\,\partial t}\,\Delta v\,\Delta t + \frac{1}{2}\frac{\partial^2 V}{\partial t^2}\,(\Delta t)^2.\end{aligned}$$

Not every term above is discretization error, and the distinction matters. Summed over a day, the Gamma, Vanna, and Volga compensators vanish as the hedging interval shrinks—realized quadratic variation converges to the model's $\int vS^2\,dt$, and likewise for the cross and variance brackets—and the charm, veta, and time-convexity terms are of order Δt or smaller. Those six terms measure finite-mesh error: the price of rebalancing every five minutes instead of continuously. The Vega term is different. Its sum converges to $\int \frac{\partial V}{\partial v}\,\epsilon\sqrt{v}\,dW$, a random quantity that survives any rebalancing frequency. It is not discreteness at all; it is the variance risk that a delta-only hedge leaves open, because the hedge holds no second option. The Heston derivation above removed this risk with the hedging option V_1; the experiment below deliberately holds only the stock, as most 0DTE hedgers do, so the Vega term tracks genuinely unhedged risk rather than finite-mesh error. The decomposition therefore separates two kinds of leftovers: terms that shrink with the hedging interval, and one term that no hedging interval can shrink.

3.2 Methodology

The Heston parameters were not calibrated to data: apart from η, they are round, arbitrarily chosen values. η was set from the VIX-implied variance in the sample. Given that information, the results in the next section are not to be taken as absolute, but rather as a proof of concept. The parameters used are as follows:

- $\kappa = 3.0$ (rate of mean reversion)
- $\eta = 0.04$ (long-run variance, the average VIX-implied variance $\overline{(\text{VIX}/100)^2} \approx 0.035$ in the sample, rounded)
- $\epsilon = 0.5$ (volatility of volatility)
- $\rho = -0.7$ (correlation between asset and variance)
- $\lambda = 0$ (market price of volatility risk)

SPX data was used as a proxy for the underlying asset. The sample covers 56 trading days, from March 31st, 2026 through June 18th, 2026. The option prices were computed using the Heston model, and the hedging error was calculated by simulating discrete hedging at various time intervals. The contributions of each Greek to the overall hedging error were then analyzed and their results are found in the following section. Better data would use actual option pricing instead of proxies, but option data is expensive and currently a limitation for this paper.

In terms of the derivation itself, the specific terms above were chosen to allow for all the terms to be expressed as Greeks, and have a discrete and nondiscrete component. The discrete component is the difference between the actual change in the underlying asset and the expected change, while the nondiscrete component is the expected change itself. The discrete component, following Bergomi's 2016 book, is what contributes to the hedging error, while the nondiscrete component is what is expected to be hedged away. If the limit is taken, the model collapses back to the Heston model for every term except Vega: the other compensators go to zero with the hedging interval, while the Vega term converges to the unhedged variance risk described above.

In the implementation, the trapezoid rule was used to compute the integrals in the Heston pricer, and the Greeks were computed using finite differences. The code is available in the accompanying Python file on GitHub. The ω grid begins at 10^{-8} to avoid the $1/(i\omega)$ singularity at zero, and the upper limit of the integral is set to 2500 to ensure convergence for 0DTE options. The number of points in the grid is set to 2048, which provides a good balance between accuracy and computational efficiency. If using a stronger computer, the number of points can be increased to improve accuracy, but at the cost of longer computation time. With testing, it has been found that if λ is set to 0, then the results are not sensitive to the number of points in the grid, as the integrals converge quickly. However, if λ is nonzero, then the results can be sensitive to the number of points in the grid, as the integrals converge more slowly. Also, setting λ to a nonzero value can introduce numerical instability in the integrals, as the integrands can become highly oscillatory, therefore making the pricer and the vega compensator disagree. For the simulation, the Euler–Maruyama method was used to simulate and calculate the stochastic differential equation used in the Heston model and the subsequent error model. The price is used as levels rather than as log prices. This was done because an Euler step $\Delta S = rS\,\Delta t + \sqrt{v}\,S\,\Delta X$ has conditional variance $vS^2\,\Delta t$, which is exactly the quantity the Gamma compensator $(\Delta S)^2 - vS^2\,\Delta t$ subtracts. Simulating in levels therefore keeps that compensator exact, so any residual left in the test reflects genuine second-order Taylor truncation rather than a mismatch between how the paths were generated and how the error was decomposed. A log-Euler step would reintroduce such a mismatch through the $-\frac{1}{2}v$ Itô correction, which the level scheme does not carry.

The variance path is read off the VIX, which is a proxy in two respects. First, VIX^2 is a risk-neutral quantity rather than a physical one, and it embeds a variance risk premium that the model, with $\lambda = 0$, does not capture. Second, and more consequentially, VIX^2 is the expected *average* variance over the coming thirty days, whereas the model calls for the instantaneous spot variance v_t. Because variance mean-reverts, a thirty-day forward average is not the same object as the spot value—it is pulled toward the long-run level and barely moves intraday, which is part of why the variance terms are only lightly exercised on real data. Cboe's VIX1D index, introduced in 2023 and computed from zero- to one-day SPXW options [12], is the horizon-matched measure and should replace the VIX in future work. For the proof of concept here, the VIX remains a reasonable stand-in.

3.3 Validation

According to Bertsimas, Kogan, and Lo (2000) [7], the granularity constant g—the measure of how difficult an option is to hedge in discrete time, and the constant that sets the size of the tracking error—blows up in the terminal interval. Along with that, the Taylor expansion fails as the option approaches expiry, causing the 9% residual error reported in the next subsection. To illustrate this, a 60-day option was simulated under the same Heston engine and hedged daily, with the position closed 20 days before expiry; there the residual was only 2.35%, against the 9% at 0DTE. The comparison is suggestive but uncontrolled: the two tests differ in data source (simulated versus market), maturity, hedge frequency, and distance from expiry at close, so the gap cannot be attributed to the terminal interval alone.

3.4 Results and Testing

The decomposition was tested on 56 full trading days of five-minute data (78 bars per day, so 77 hedge intervals). Each day is rescaled to open at 1, the variance path is read off the squared VIX, and the option is a fresh at-the-money 0DTE call re-formed each morning and delta-hedged at every bar. Summing the seven compensated terms reproduces the realized hedged P&L to within a 9% residual—defined as the mean absolute unexplained P&L divided by the mean absolute daily P&L—with a mean daily hedged P&L of -0.00235 (in the units where the underlying opens at 1) and a standard deviation of 0.00122.

Table 1 attributes that P&L term by term. The percentages are gross absolute model-attribution shares: each term's average absolute daily contribution as a fraction of the sum across all seven terms, computed from model Greeks. Because terms can offset within a day, these shares measure how much each term moves, not a split of the realized hedging error. In the table, Vega, Vanna, Volga, and Veta denote derivatives with respect to the variance v, not the volatility σ of Section 1; the two are related by chain rule but are not numerically identical. The shares are reported first over all days, then split into three regimes by how far the VIX travelled intraday.

Term	Overall	Calm	Mid	Turbulent
Gamma $\left(\frac{1}{2}\,\partial^2 V/\partial S^2\right)$	89.74	93.55	91.54	82.11
Vega $(\partial V/\partial v)$	4.79	1.25	3.53	11.33
Vanna $(\partial^2 V/\partial S\,\partial v)$	2.35	2.19	2.00	3.01
Charm $(\partial^2 V/\partial S\,\partial t)$	1.50	1.53	1.33	1.70
Time convexity $\left(\frac{1}{2}\,\partial^2 V/\partial t^2\right)$	1.27	1.16	1.29	1.39
Volga $\left(\frac{1}{2}\,\partial^2 V/\partial v^2\right)$	0.25	0.27	0.23	0.23
Veta $(\partial^2 V/\partial v\,\partial t)$	0.11	0.04	0.08	0.23

Table 1: Gross absolute model-attribution share of each decomposition term, as a percentage of the summed average absolute daily contributions. Days are sorted into terciles by the absolute intraday change in the VIX: calm ($n = 19$, $\langle|\Delta\text{VIX}|\rangle = 0.12$), mid ($n = 18$, 0.47), and turbulent ($n = 19$, 1.57).

Gamma dominates in every regime, as one would expect of an option hedged only in the underlying: on a typical day roughly nine-tenths of the hedging error is the $\frac{1}{2}\,\partial^2 V/\partial S^2$ term. What shifts with the regime is where the remaining tenth comes from. On calm days Gamma carries 94% and Vega barely 1%; on turbulent days Gamma's share falls to 82% while Vega climbs to 11%, absorbing almost all of the share Gamma gives up. Part of this pattern is mechanical—sorting days by $|\Delta\text{VIX}|$ necessarily favors the Vega term in the turbulent bucket—so the informative facts are the size of the swing and Gamma's persistence, not the direction of the shift. The volatility-driven error is essentially dormant when the VIX sits still and becomes the second-order story precisely when it does not. Everything else—Vanna, Charm, time convexity, Volga, and Veta—stays within a few percent throughout, confirming that for 0DTE the hedging error lives overwhelmingly in Gamma, with Vega as the regime-dependent runner-up.

3.5 Limitations and Future Work

The model is limited by the accuracy of the Heston parameters, an issue one can easily fix with better data. Another limitation is found in the premise itself: all Greeks are unhedgeable in the terminal interval, because there exists no rebalance prior to expiry. Given that, firms, market makers, retail traders, and the like should be wary in their usage of 0DTE options, as they are inherently risky and difficult to hedge, especially as they near expiration. This paper simply explores the sources of where the hedging error goes, and does not provide a definitive solution to the problem. To make this model better, a Milstein scheme [4] could be used to simulate the variance process, which would provide a more accurate representation of the underlying asset's volatility.

A further limitation is structural. Heston is a pure-diffusion model, so the decomposition carries no jump term; any intraday jump in the underlying enters only through $(\Delta S)^2$ and is therefore absorbed into the Gamma term, indistinguishable from ordinary diffusive movement. This matters for 0DTE in particular: Božović (2025) [11], working with five-minute SPXW options, estimates the implied jump-risk premium at nearly twice the combined diffusion and volatility premia, with jump intensity clustering at the open and close. The roughly ninety-percent Gamma share reported above should therefore be read as diffusion and jump risk combined; isolating the two would require adding a jump term to the decomposition, a natural extension for future work.

4 Conclusion

Learning the different attributions of the error is important for understanding the risks of 0DTE options and being able to use them in an effective manner. The results show that the majority of the error is carried by Gamma, which the model conflates with jumps. The remaining error is mostly regime-dependent, as Vega—the variance risk a delta-only hedge cannot remove at any rebalancing frequency—increases its contribution as the regime becomes increasingly turbulent. The other Greeks, while not negligible, are relatively small and do not change much across regimes. The results of this paper are a proof of concept, and further work is needed to refine the model and improve the accuracy of the results. Future work should focus on improving the Heston parameter estimates, adding a jump term to the decomposition, and testing the model on a wider range of data.

A Glossary of Variables

The sensitivity Greeks (Δ, Γ, ν, Θ, and the higher-order terms) are defined in Section 1. The symbols below collect the model, market, and hedging notation that the Heston and derivation sections introduce along the way.

Underlying and market.

- S: price of the underlying asset (the spot), rescaled so that each trading day opens at 1.
- v: instantaneous variance of the underlying; the volatility is $\sqrt{v}$.
- σ: volatility, with $\sigma = \sqrt{v}$; used in the Greek definitions of Section 1.
- t: calendar time. Written as a prefix, Δt is a single hedging interval (one five-minute bar in the test).
- μ: real-world (physical) drift of the underlying.
- r: risk-free interest rate ($r = 0.045$ in the test).
- X, W: the Brownian motions driving the price and the variance, correlated by $dX\,dW = \rho\,dt$.

Heston parameters.

- κ: rate at which the variance reverts toward its long-run level.
- η: long-run (mean) level of the variance.
- ϵ: volatility of volatility.
- ρ: correlation between the price and variance shocks.
- λ: market price of volatility risk; $\lambda = 0$ in the test, so the physical and risk-neutral variance drifts coincide.
- ν_{Q}: risk-neutral drift of the variance, $\nu_{\mathrm{Q}} = \kappa(\eta - v) - \lambda v$ (distinct from Vega ν).

Option and hedge portfolio.

- V: value of the option being hedged (a fresh at-the-money 0DTE call).
- V_1: value of the second option held to hedge variance (Vega) risk.
- Π: value of the delta–vega hedged portfolio, $\Pi = V - \Delta S - \Delta_1 V_1$.
- Δ: units of the underlying held in the hedge, $\Delta = \partial V/\partial S - \Delta_1\,\partial V_1/\partial S$. Written as a prefix ($\Delta S$, Δv, $\Delta\Pi$) it instead denotes the change in a quantity over one hedging interval.
- Δ_1: units of the hedging option V_1 held, chosen to cancel the portfolio's net Vega, $\Delta_1 = (\partial V/\partial v)/(\partial V_1/\partial v)$.

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